



Cinvestav
Guadalajara



COCYTEN
CONSEJO DE CIENCIA Y TECNOLOGÍA
DEL ESTADO DE NAYARIT



Nayarit
NUESTRA DIGNIDAD Y COMPROMISO



Talento Altamente Especializado – Inteligencia Artificial **2025 COCYTEN-Nayarit**

Lab 1: Python & Tools Installation



Students: Ricardo Torres López y Cesar Eduardo Inda Cenicerros

Instructor: Dr. Cristian Torres González

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Submission Date: 10/10/2025



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Introduction

To continue with the course, it is important to have the necessary tools installed and complete the following activities. Tasks marked as (Urgent) must be completed before the second class.

Laboratory Description

Complete Questionnaire:

<https://forms.gle/WWzMHrpqe6kZxcZf8> (Done!!)

Microsoft Documents → General. Register the Team in the Table.

Documentos > General			
Nombre	Modificado	Modificado p...	
Labs	hace 18 minutos	Cristian Torres Go	
Recordings	Hace una hora	Aplicación de Sha	
Slides	hace 21 minutos	Cristian Torres Go	
Teams	hace 18 minutos	Cristian Torres Go	

Team	Members	Phone number	Email
0	Cristian Torres Gonzalez Cristofer Paredes	5519641626 4778171440	Cristian.Torres@cinvestav.mx
1			
2			

	Name	Phone number	Email
0			
1			
2			

1.- Install Anaconda (Navigator)

Sign in for AI Coding Help

Create a function to mask email addresses, show examples (using no context from Notebook)

To create a function called "mask_email" that takes an email address as input:

1. Define a function called "mask_email" that takes an email address as input.
2. Split the email address into two parts: the username and the domain.
3. Replace all characters in the username except for the first and last character with asterisks.
4. Return the masked email address by concatenating the masked username with the domain.

Here's an example implementation of the "mask_email" function:

```
def mask_email(email):
    username, domain = email.split('@')
    masked_username = '*' + username[1:-1] + '*'
    masked_email = masked_username + '@' + domain
    return masked_email
```

Here are some examples of how you can use the "mask_email" function:

Reply to this chat...

Access Anaconda Assistant in Jupyter Notebooks and JupyterLab.

☐ Do not show again

[Connect to a Repository](#) [Sign In](#)

Don't have an account? [Sign Up](#)



2.- Run “Hello World” program in Anaconda Prompt (Anaconda)

```
Administrator: Anaconda Prompt
Type "help", "copyright", "credits" or "license" for more information.
>>> print("This is Anaconda Prompt... This is for test.. Cesar Inda!")
This is Anaconda Prompt... This is for test.. Cesar Inda!
>>> quit()

(base) C:\WINDOWS\system32>python -m this
The Zen of Python, by Tim Peters

Beautiful is better than ugly.
Explicit is better than implicit.
Simple is better than complex.
Complex is better than complicated.
Flat is better than nested.
Sparse is better than dense.
Readability counts.
Special cases aren't special enough to break the rules.
Although practicality beats purity.
Errors should never pass silently.
Unless explicitly silenced.
In the face of ambiguity, refuse the temptation to guess.
There should be one-- and preferably only one --obvious way to do it.
Although that way may not be obvious at first unless you're Dutch.
Now is better than never.
Although never is often better than *right* now.
If the implementation is hard to explain, it's a bad idea.
If the implementation is easy to explain, it may be a good idea.
Namespaces are one honking great idea -- let's do more of those!

(base) C:\Users\dell>python
Python 3.13.5 | packaged by Anaconda, Inc. | (main, Jun 12 2025, 16:37
:03) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print("This is Ricardo, Hello World!!!")
This is Ricardo, Hello World!!!
>>> quit()

(base) C:\Users\dell>
```

Create a presentation slide that includes:

Cesar Eduardo Inda Cenicerós

- Cybersecurity Engineer and Mechatronic Engineer
- Specialization Embedded Systems
- Interest: Electronic And Semiconductors Security.
- 2 years exp. Embedded Software Security Developer (Automotive)
- 1 year exp. Embedded Software Security Developer (Payment Devices)

Ricardo Torres López

- Chemical Engineer
- Designer in Unit Operations.
- Electronic and semiconductor
- 12 years as Director
- 5 years as a trader (yellow belt).

```
PS D:\Cinvestav\Modulo1_IA\PythonCourse2025> & C:/ProgramData/anaconda3/python.exe d:/Cinvestav/Modulo1_IA/PythonCourse2025/program.py
Hello, World!
PS D:\Cinvestav\Modulo1_IA\PythonCourse2025>
```



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d) Jupyter Notebook

We ran the program in the “**Visual Studio Code**” extension. We execute the code and then execute the command in the terminal “**Jupyter notebook - -port=8080**”.

```
[I 2025-10-09 16:51:40.617 ServerApp] Serving notebooks from local directory: D:\Cinvestav\Modulo1_IA\PythonCourse2025
[I 2025-10-09 16:51:40.617 ServerApp] Jupyter Server 2.15.0 is running at:
[I 2025-10-09 16:51:40.618 ServerApp] http://localhost:8080/tree?token=ad17ceb76d156f699246887aa5b3e5a9ce93318734ea982c
[I 2025-10-09 16:51:40.618 ServerApp] http://127.0.0.1:8080/tree?token=ad17ceb76d156f699246887aa5b3e5a9ce93318734ea982c
[I 2025-10-09 16:51:40.619 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 2025-10-09 16:51:40.784 ServerApp]

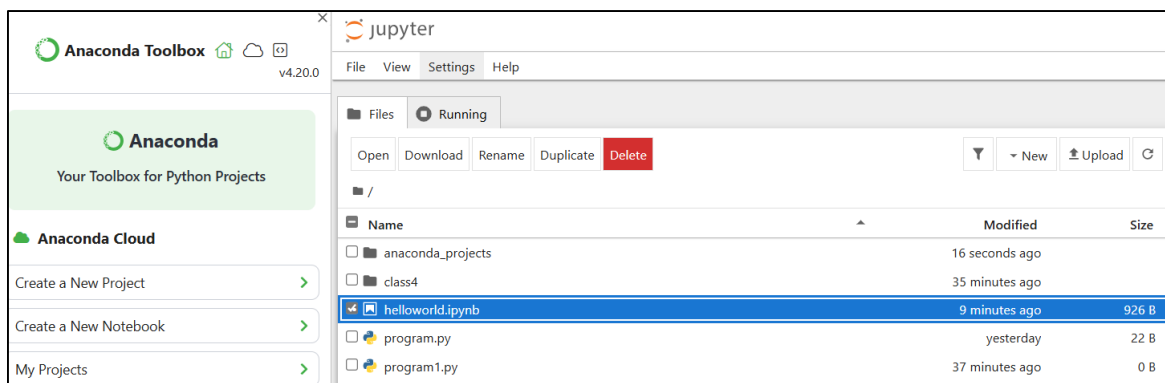
To access the server, open this file in a browser:
file:///C:/Users/tiori/AppData/Roaming/jupyter/runtime/jpserver-6744-open.html
Or copy and paste one of these URLs:
http://localhost:8080/tree?token=ad17ceb76d156f699246887aa5b3e5a9ce93318734ea982c
http://127.0.0.1:8080/tree?token=ad17ceb76d156f699246887aa5b3e5a9ce93318734ea982c
[I 2025-10-09 16:51:41.326 ServerApp] Skipped non-installed server(s): bash-language-server, dockerfile-language-server-nodejs, javascript-typescript-langserver, jedi-language-server, python-language-server, r-languageserver, sql-language-server, texlab, typescript-language-server, unified-language-server, vscode-css-languageserver-bin, vscode-html-languageserver-bin, yaml-language-server
```

Then we can see the connection to kernel.

```
CONDA_PREFIX=C:\ProgramData\anaconda3
[I 2025-10-09 16:52:41.930 ServerApp] Connecting to kernel ab991b64-3979-43ec-bbab-c0da8249d5de.
[I 2025-10-09 16:52:41.938 ServerApp] Connecting to kernel ab991b64-3979-43ec-bbab-c0da8249d5de.
[W 2025-10-09 16:52:41.950 ServerApp] The websocket_ping_timeout (90000) cannot be longer than the websocket_ping_interval (30000).
Setting websocket_ping_timeout=30000
[I 2025-10-09 16:52:41.964 ServerApp] Connecting to kernel ab991b64-3979-43ec-bbab-c0da8249d5de.
[W 2025-10-09 16:52:42.087 ServerApp] Got events for closed stream <zmq.eventloop.zmqstream.ZMQStream object at 0x0000025658491E50>
WebSocket connection closed.
[I 2025-10-09 16:54:18.204 ServerApp] Saving file at /helloworld.ipynb
```

URL that was generated: <http://localhost:8080/notebooks/helloworld.ipynb>?

Home → Anaconda Toolbox





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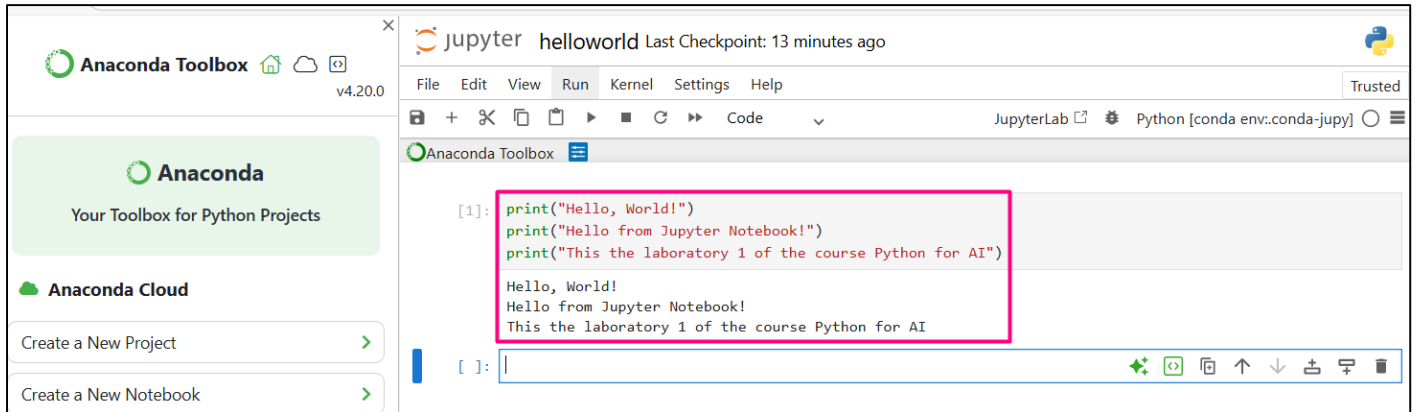
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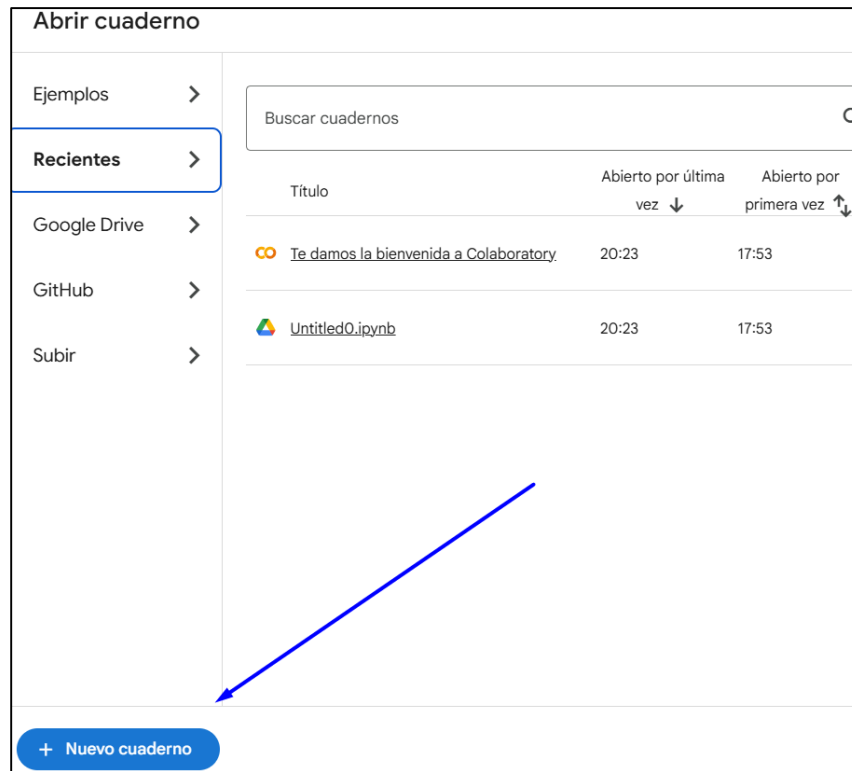
We can observe in the other windows the example of the code.



e) Google Colab

Before writing code in Google Colab, we need to install it and then open a new notebook to work with these collaborative tools. Then open the notebook.

<https://colab.google/>





After that, we run a test code using Python libraries with Jupyter Notebook extension (.ipynb)

```
Untitled0.ipynb ☆ ☁
Archivo Editar Ver Insertar Entorno de ejecución Herramientas
Comandos + Código + Texto ▶ Ejecutar todas
Empieza a programar o a crear código con IA

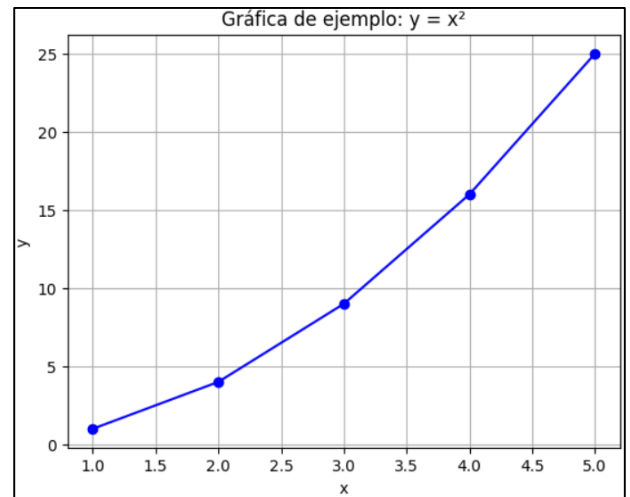
[4]
import math
import matplotlib.pyplot as plt

print("👋 ¡Hola desde Google Colab!")
print(f"El valor de  $\pi$  (pi) es: {math.pi}")

a = 10
b = 5
suma = a + b
print(f"La suma de {a} + {b} es {suma}")

x = [1, 2, 3, 4, 5]
y = [n**2 for n in x]
plt.plot(x, y, marker='o', color='blue')
plt.title("Gráfica de ejemplo:  $y = x^2$ ")
plt.xlabel("x")
plt.ylabel("y")
plt.grid(True)
plt.show()
```

👋 ¡Hola desde Google Colab!
El valor de π (pi) es: 3.141592653589793
La suma de 10 + 5 es 15





Conclusions / Issues

Cesar Eduardo Inda Cenicerros

This practice was interesting; personally, I had never used Anaconda before, nor had I explored virtual environments in such depth. For me, it was something very valuable, and I also believe that many of these tools provide a new perspective on each of the elements of the certification and the learning modality.

Ricardo Torres Lopez

As a student (beginner), it has been an amazing trip, where I learned to install, activate, deactivate, deinstall, print, etc. I faced some problems during the process of learning, nevertheless, it could be solved by my professors' instructions and supporting.

Troubleshooting

```

Instale la versión más reciente de PowerShell para obtener nuevas características y mejoras. https://aka.ms/PSWindows

. : No se puede cargar el archivo C:\Users\dell\Documents\WindowsPowerShell\profile.ps1 porque la ejecución de scripts
está deshabilitada en este sistema. Para obtener más información, consulta el tema about_Execution_Policies en
https://go.microsoft.com/fwlink/?LinkID=135170.
En línea: 1 Carácter: 3
+ . 'C:\Users\dell\Documents\WindowsPowerShell\profile.ps1'
+ ~~~~~
+ CategoryInfo          : SecurityError: (:) [], PSSecurityException
+ FullyQualifiedErrorId : UnauthorizedAccess

PS C:\Windows\system32> conda upgrade conda
3 channel Terms of Service accepted
Channels:
- defaults
Platform: win-64
Collecting package metadata (repodata.json): done
Solving environment: done
  
```

Before solving the problem, we must understand the differences between Anaconda prompt and Anaconda PowerShell prompt.

Simple Explanation:

Both terminals do the same main job:

- Run Conda commands
- Manage Python environments
- Install packages

Main differences:

Anaconda Prompt	Anaconda PowerShell Prompt
Older style	Newer style
Simple commands	Smart commands
Works with old scripts	More modern features

Which to choose?

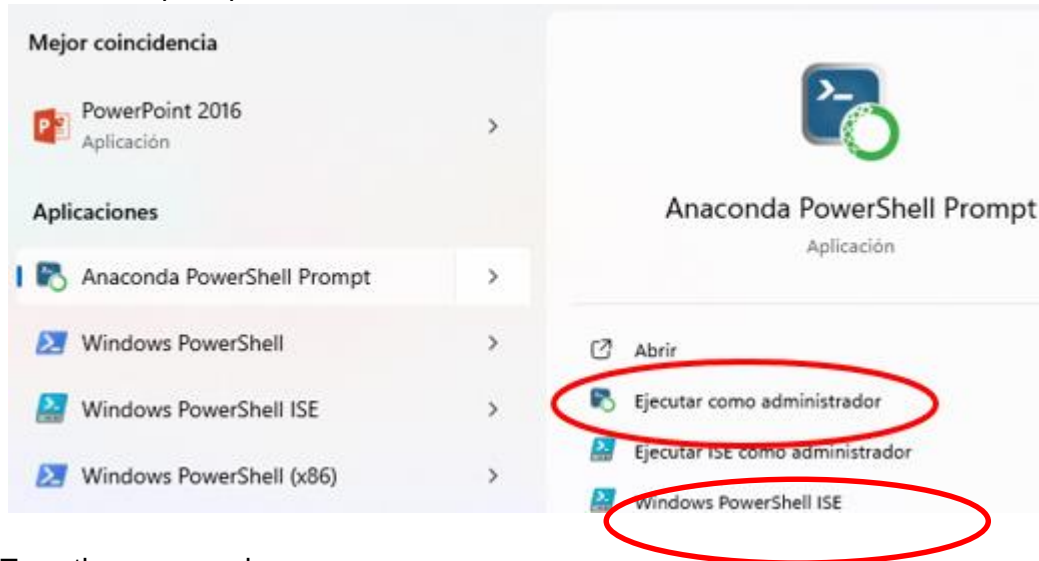
- If you want **simple and familiar** → Anaconda Prompt
- If you want **modern and powerful** → Anaconda PowerShell Prompt

Remember: Both can use all Conda commands like `conda install`, `conda create`, etc.



Therefore, the best option will be Anaconda PowerShell prompt to solve the troubleshot generated for the system. Follow the next steps:

1. Activate the prompt as a administrator



2. Type the command

```
Administrador: Anaconda PowerShell Prompt
(base) PS C:\Windows\system32> Get-ExecutionPolicy -List

Scope ExecutionPolicy
-----
MachinePolicy Undefined
UserPolicy Undefined
Process Bypass
CurrentUser Unrestricted
LocalMachine Unrestricted
```

3. Then type **Set-ExecutionPolicy -ExecutionPolicy Unrestricted -Scope LocalMachine**
4. Finally, type **Set-ExecutionPolicy -ExecutionPolicy Unrestricted -Scope CurrentUser**

GitHub Repository

<https://github.com/CeicGitHub/Specialization Artificial Intelligence Module1/tree/Lab1 Python AndTools>

<https://github.com/RicardoTL75/Modulo1 AI/tree/Module1 Python Tools>